

Week 6

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1. Useful Properties

Property 1

Let A be an $m \times n$ matrix. Then, the eigenvalues of $A^T A$ and AA^T are greater than or equal to 0.

Proof

Let $(A^T A)u = \lambda_1 u$ and let $(AA^T)v = \lambda_2 v$. Then

$$\begin{aligned} (A^T A)u &= \lambda_1 u & (AA^T)v &= \lambda_2 v \\ u^T(A^T A)u &= \lambda_1 u^T u & v^T(AA^T)v &= \lambda_2 v^T v \\ (Au)^T Au &= \lambda_1 ||u||^2 & (A^T v)^T A^T v &= \lambda_2 ||v||^2 \\ ||Au||^2 &= \lambda_1 ||u||^2 & ||A^T v||^2 &= \lambda_2 ||v||^2 \\ \lambda_1 &= \frac{||Au||^2}{||u||^2} & \lambda_2 &= \frac{||Av||^2}{||v||^2} \end{aligned}$$

As u and v are eigenvectors of $A^T A$ and AA^T respectively, they are non-zero vectors. Therefore, $||u||^2 \neq 0$ and $||v||^2 \neq 0$.

Also,

$$\lambda_1 = \frac{||Au||^2}{||u||^2} \geq 0 \qquad \lambda_2 = \frac{||Av||^2}{||v||^2} \geq 0$$

Property 2

$A_{m \times n}$ and $B_{n \times m}$ are two matrices. Let $P_{n \times n}$ is an invertible matrix. Then, $\text{rank}(AP) = \text{rank}(A)$ and $\text{rank}(PB) = \text{rank}(B)$

Proof

For any two matrices X and Y , the following is true.

$$\text{rank}(XY) \leq \min(\text{rank}(X), \text{rank}(Y))$$

Since P is an invertible matrix,

$$\begin{aligned} \text{rank}(AP) &\leq \min(\text{rank}(A), \text{rank}(P)) \\ &\leq \text{rank}(A) \end{aligned}$$

Also,

$$\begin{aligned} \text{rank}(APP^{-1}) &\leq \min(\text{rank}(AP), \text{rank}(P^{-1})) \\ &\leq \text{rank}(AP) \end{aligned}$$

Note : $APP^{-1} = A$. From the above two inequalities, we get :

$$\text{rank}(A) \leq \text{rank}(AP) \leq \text{rank}(A)$$

The above inequality is true only if $\text{rank}(AP) = \text{rank}(A)$. Similarly, we can show that $\text{rank}(PB) = \text{rank}(B)$

Property 3

Suppose $A = PBP^{-1}$. Then, $\text{rank}(A) = \text{rank}(B)$.

Proof

$$\begin{aligned} A &= PBP^{-1} \\ AP &= PB \\ \text{rank}(AP) &= \text{rank}(PB) \\ \text{rank}(A) &= \text{rank}(B) \end{aligned}$$

The rank of similar matrices are equal.

Property 4

Let $A_{n \times n}$ be a diagonalizable matrix. That is, $A = PDP^{-1}$, where P is an invertible matrix and D is a diagonal matrix. The rank of A is equal to the number of non-zero eigenvalues of A .

Proof

We know that the rank of similar matrices are equal.

$$\text{Rank}(A) = \text{Rank}(D)$$

Rank of a diagonal matrix is equal to the number of non-zero diagonal entries. The diagonal entries of a diagonal matrix are its eigenvalues. Moreover, similar matrices have the same eigenvalues. Therefore, rank of A is the number of non-zero eigenvalues of A .

Note that the eigenvalues can repeat. That is, if A is a diagonalizable matrix with 1, 2, 2 and 0 as its eigenvalues, then the rank of A is 3. Note that the eigenvalue 2 is repeated twice.

Property 5

Let A and B be two invertible matrices. Suppose, $A = PBP^{-1}$, i.e., A and B are similar. Then, A^{-1} and B^{-1} are also similar.

Proof

$$\begin{aligned} A &= PBP^{-1} \\ A^{-1} &= (PBP^{-1})^{-1} \\ A^{-1} &= (P^{-1})^{-1}B^{-1}P^{-1} \\ A^{-1} &= PB^{-1}P^{-1} \end{aligned}$$

We have used the following fact:

$$(MN)^{-1} = N^{-1}M^{-1}$$

Property 6

Let A be an $m \times n$ matrix. Then,

$$\text{Rank}(A) = \text{Rank}(A^T A) = \text{Rank}(AA^T)$$

Proof

First, we shall prove $\text{Rank}(A) = \text{Rank}(A^T A)$.

If $v \in N(A)$, then $v \in N(A^T A)$.

$$Av = 0$$

$$A^T A v = A^T 0$$

$$A^T A v = 0$$

If $v \in N(A^T A)$, then $v \in N(A)$

$$A^T A v = 0$$

$$v^T A^T A v = v^T 0$$

$$(A v)^T A v = 0$$

$$||A v||^2 = 0$$

$$A v = 0$$

Therefore, $N(A) = N(A^T A)$. This means, $nullity(A) = nullity(A^T A)$. Both, A and $A^T A$ have n columns. Using **rank-nullity** theorem, we get the following.

$$Rank(A) + Nullity(A) = n$$

$$Rank(A) = n - Nullity(A) \quad (1)$$

$$Rank(A^T A) + Nullity(A^T A) = n$$

$$Rank(A^T A) = n - Nullity(A^T A) \quad (2)$$

RHS of (1) and (2) are equal, therefore,

$$Rank(A) = Rank(A^T A)$$

Let $B = A^T$. Using the above equation, we get

$$Rank(B) = Rank(B^T B)$$

$$Rank(A^T) = Rank(AA^T)$$

$$Rank(A) = Rank(AA^T)$$

We have used the following fact

$$Rank(A) = Rank(A^T)$$

Property 7

If λ_1 is an eigenvalue of $A^T A$, then λ_1 is also an eigenvalue of AA^T . Similarly, if λ_2 is an eigenvalue of AA^T , then λ_2 is also an eigenvalue of $A^T A$.

Proof

Let $A^T A u = \lambda_1 u$ and $AA^T v = \lambda_2 v$.

$$\begin{aligned}
A^T A u &= \lambda_1 u & A A^T v &= \lambda_2 v \\
A A^T A u &= \lambda_1 A u & A^T A A^T v &= \lambda_2 A^T v \\
A A^T (A u) &= \lambda_1 (A u) & A^T A (A^T v) &= \lambda_2 (A^T v)
\end{aligned}$$

We have proved that the property is true. There is something more that we can conclude from the above property.

$A^T A$ and $A A^T$ are symmetric, and therefore, are diagonalizable. This means, the number of non-zero eigenvalues of $A^T A$ is its rank. Similarly, the number of non-zero eigenvalues of $A A^T$ is equal to its rank.

Since $\text{Rank}(A^T A) = \text{Rank}(A A^T)$, we see that the number of non-zero eigenvalues of $A^T A$ and $A A^T$ are equal. Since we have proved the above property, we can additionally say that the non-zero eigenvalues of $A^T A$ and $A A^T$ are the same.

Property 8

Let A be an $m \times n$ matrix. The eigenvectors corresponding to the non-zero eigenvalues of $A^T A$ form a basis for the row space of A .

Proof

First,

$$\text{Rank}(A) = \text{Rank}(A^T A)$$

Also, $\text{Rank}(A) = \text{Row Rank}(A)$. The rank of $A^T A$ is the number of non-zero eigenvalues of $A^T A$. Which means, the number of non-zero eigenvalues of $A^T A$ is equal to the row rank of A .

Suppose, $\text{Rank}(A) = r = \text{Rank}(A^T A)$, where $r \leq \min(m, n)$. Clearly, r is also less than or equal to n . This is because, $A^T A$ has n rows and n columns.

Let $V = \{v_1, \dots, v_r\}$ be the eigenvectors corresponding to the non-zero eigenvalues of r . As $A^T A$ is diagonalizable, we can choose $\{v_1, \dots, v_r\}$ to be a linearly independent set. Recall, $A^T A$ has r non-zero eigenvalues.

$$A^T A v_i = \lambda_i v_i, \quad i \in \{1, 2, \dots, r\}$$

Notice, $A^T A v_i$ is a linear combination of the columns of A^T . This means,

$$A^T A v_i \in C(A^T)$$

where, $C(A^T)$ denotes the column space of A^T . The column space of A^T is the row space of A . Therefore,

$$A^T A v_i \in R(A)$$

$$\lambda_i v_i \in R(A)$$

$$v_i \in R(A)$$

Here, $R(A)$ denotes the row space of A . If a vector is an element of a vector space, then any scalar multiple of that vector is also an element in that vector space. Since $\lambda_i v_i \in R(A)$, we get $v_i \in R(A)$.

Clearly, every eigenvector v_i , (where $i \in \{1, \dots, r\}$) is an element of $R(A)$. The set $V = \{v_1, \dots, v_r\}$ is a linearly independent set and $V \subseteq R(A)$. The rank of A is r . The set V also has r linearly independent eigenvectors of $A^T A$. Therefore,

$$\text{span}(V) = R(A)$$

That is, every vector in the row space of A is a linear combination of the eigenvectors of $A^T A$ that correspond to the non-zero eigenvalues of $A^T A$.

Property 9

Let $Q_{n \times n}$ be an orthogonal matrix. Let $u, v \in \mathbb{R}^n$. Let θ° be the angle between u and v . Then,

- $\|u\|^2 = \|Qu\|^2$
- The angle between Qu and Qv is θ°

Proof

$$\begin{aligned} \|Qu\|^2 &= (Qu)^T (Qu) \\ &= u^T Q^T Q u \\ &= u^T I u \\ &= \|u\|^2 \end{aligned}$$

When a vector is multiplied with an orthogonal matrix, the length is preserved.

Since θ° is the angle between u and v , the following is true.

$$\cos(\theta^\circ) = \frac{u^T v}{\|u\| \times \|v\|}$$

Let α° be the angle between Qu and Qv . Then

$$\begin{aligned}\cos(\alpha^\circ) &= \frac{(Qu)^T(Qv)}{\|Qu\| \times \|Qv\|} \\&= \frac{u^T Q^T Q v}{\|u\| \times \|v\|} \\&= \frac{u^T v}{\|u\| \times \|v\|} \\&= \cos(\theta^\circ) \\ \alpha^\circ &= \theta^\circ\end{aligned}$$

When two vectors are multiplied by an orthogonal matrix, the angle between the two vectors is preserved.

2. Singular Value Decomposition

Theorem : Let A be an $m \times n$ matrix and let $\text{Rank}(A) = r$, where $r \leq \min(m, n)$. Matrix A can be decomposed as follows

$$A = U\Sigma V^T$$

where,

- U is an $m \times r$ matrix with orthonormal columns
- V is an $n \times r$ matrix with orthonormal columns
- Σ is a $r \times r$ diagonal matrix

Proof :

$A^T A$ is an $n \times n$ matrix and AA^T is a $m \times m$ matrix. The rank of these two matrices is r . Both have the same non-zero eigenvalues. They are :

$$\lambda_1, \dots, \lambda_r$$

Here, $\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_r > 0$

$A^T A$ and AA^T are symmetric, therefore, can have orthonormal set of eigenvectors.

Consider matrix V and U

$$V = \begin{bmatrix} | & | & \cdots & | \\ v_1 & v_2 & & v_r \\ | & | & & | \end{bmatrix}_{n \times r}, \quad U = \begin{bmatrix} | & | & \cdots & | \\ u_1 & u_2 & & u_r \\ | & | & & | \end{bmatrix}_{m \times r}$$

Columns of V are orthonormal. They are the eigenvectors of $A^T A$. v_i corresponds to λ_i , where $i \in \{1, 2, \dots, r\}$.

$$A^T A v_i = \lambda_i v_i$$

Columns of U are also orthonormal. They are the eigenvectors of AA^T . u_i corresponds to λ_i , where $i \in \{1, 2, \dots, r\}$.

$$AA^T u_i = \lambda_i u_i$$

Let us focus on $A^T A v_i = \lambda_i v_i$

$$A^T A v_i = \lambda_i v_i$$

$$AA^T A v_i = \lambda_i A v_i$$

If v_i is an eigenvector of $A^T A$ with λ_i as the eigenvalue, then $A v_i$ is the eigenvector of AA^T with λ_i as the eigenvalue. We know that u_i is also an eigenvector of AA^T with λ_i as the eigenvalue. However, $\|u_i\| = 1$. If we divide $A v_i$ by $\|A v_i\|$, we obtain u_i . Let us find $\|A v_i\|$

$$\begin{aligned} \|A v_i\|^2 &= (A v_i)^T (A v_i) \\ &= v_i^T A^T A v_i \\ &= v_i^T (\lambda_i v_i) \\ &= \lambda_i (v_i^T v_i) \\ &= \lambda_i \\ \|A v_i\| &= \sqrt{\lambda_i} \end{aligned}$$

We have used the fact that $\|v_i\| = 1$. Let us denote $\sigma_i = \sqrt{\lambda_i}$.

$$u_i = \frac{A v_i}{\|A v_i\|} = \frac{A v_i}{\sqrt{\lambda_i}} = \frac{A v_i}{\sigma_i}$$

We now get the following equation.

$$A v_i = \sigma_i u_i$$

This is true for all $i \in \{1, 2, \dots, r\}$. Therefore, we get the following equations

$$\begin{aligned}
Av_1 &= \sigma_1 u_1 \\
Av_2 &= \sigma_2 u_2 \\
&\vdots \\
Av_i &= \sigma_r u_r
\end{aligned}$$

The matrix representation for the above system of linear equations is $AV = U\Sigma$, where

$$V = \begin{bmatrix} | & | & & | \\ v_1 & v_2 & \cdots & v_r \\ | & | & & | \end{bmatrix}_{n \times r}, \quad U = \begin{bmatrix} | & | & & | \\ u_1 & u_2 & \cdots & u_r \\ | & | & & | \end{bmatrix}_{m \times r}, \quad \Sigma = \begin{bmatrix} \sigma_1 & 0 & \cdots & 0 \\ 0 & \sigma_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \sigma_r \end{bmatrix}_{r \times r}$$

Let us get back to $AV = U\Sigma$

$$\begin{aligned}
AV &= U\Sigma \\
AVV^T &= U\Sigma V^T \\
((VV^T)^T(A)^T)^T &= U\Sigma V^T \\
(VV^T A^T)^T &= U\Sigma V^T
\end{aligned}$$

- We have used : $(XY)^T = Y^T X^T$.
- VV^T is symmetric. Therefore, $(VV^T)^T = VV^T$

VV^T is also the projection matrix that projects all the vectors in $\mathbb{R}^n$ onto the column space of V .

$$\begin{aligned}
V(V^T V)^{-1} V^T &= V(I)^{-1} V^T \\
&= VI^{-1} V^T \\
&= VIV^T \\
&= VV^T
\end{aligned}$$

The columns of V are linearly independent. Therefore, $(V^T V)^{-1}$ exists. Moreover, we can show that $VV^T = I$, where I is the identity matrix.

$$V^T V = \begin{bmatrix} - & v_1^T & - \\ - & v_2^T & - \\ & \vdots & \\ - & v_r^T & - \end{bmatrix} \begin{bmatrix} | & | & & | \\ v_1 & v_2 & \cdots & v_r \\ | & | & & | \end{bmatrix} = \begin{bmatrix} v_1^T v_1 & v_1^T v_2 & \cdots & v_1^T v_r \\ v_2^T v_1 & v_2^T v_2 & \cdots & v_2^T v_r \\ \vdots & \vdots & \ddots & \vdots \\ v_r^T v_1 & v_r^T v_2 & \cdots & v_r^T v_r \end{bmatrix} = \begin{bmatrix} 1 & 0 & \cdots & 0 \\ 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 1 \end{bmatrix}$$

The columns of V are the eigenvectors corresponding to the non-zero eigenvalues of $A^T A$. Recall, every vector in the row space of A is a linear combination of the eigenvectors corresponding to the non-

zero eigenvalues of $A^T A$. That is, every vector in the row space is a linear combination of $\{v_1, \dots, v_r\}$. The column vectors of A^T are the row vectors of A .

VV^T is the projection matrix that projects any vector in $\mathbb{R}^n$ onto the row space of A . In general, if P is a projection matrix which projects a vector onto a vector space K , then $Pk = k$, for all $k \in K$.

Since every column vector of A^T is in the row space of A , $VV^T A^T = A^T$.

$$(VV^T A^T)^T = U \Sigma V^T$$

$$(A^T)^T = U \Sigma V^T$$

$$A = U \Sigma V^T$$

3. Full Picture of SVD

In the previous section we have shown that any matrix A can be decomposed into $A = U \Sigma V^T$, where

- the columns of $U_{m \times r}$ are orthonormal
- the columns of $V_{n \times r}$ are orthonormal
- $\Sigma_{r \times r}$ is a diagonal matrix

The above form is the **truncated** form of SVD. The **full picture** of SVD is as follows:

Theorem : Let A be an $m \times n$ matrix and let $\text{Rank}(A) = r$, where $r \leq \min(m, n)$. Matrix A can be decomposed as follows

$$A = U \Sigma V^T$$

where,

- U is an $m \times m$ orthogonal matrix
- V is an $n \times n$ orthogonal matrix
- Σ is a $m \times n$ matrix. All the elements that are not in the principal diagonal of Σ are 0.

Proof :

Let $\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_r > 0$ be the r non-zero eigenvalues of $A^T A$. Matrices V , U and Σ are given as follows

$$V = \begin{bmatrix} | & | & & | & | & & | \\ v_1 & v_2 & \cdots & v_r & v_{r+1} & \cdots & v_n \\ | & | & & | & | & & | \end{bmatrix}_{n \times n}, \quad U = \begin{bmatrix} | & | & & | & | & & | \\ u_1 & u_2 & \cdots & u_r & u_{r+1} & \cdots & u_m \\ | & | & & | & | & & | \end{bmatrix}_{m \times m},$$

$$\Sigma = \begin{bmatrix} \sigma_1 & 0 & \cdots & 0 & 0 & \cdots & 0 \\ 0 & \sigma_2 & \cdots & 0 & 0 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \sigma_r & 0 & \cdots & 0 \\ 0 & 0 & \cdots & 0 & 0 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 0 & 0 & \cdots & 0 \end{bmatrix}_{m \times n}$$

The first r entries of the principal diagonals of Σ are $\sigma_1, \dots, \sigma_r$. Here, $\sigma_i = \sqrt{\lambda_i}$.

V is an orthogonal matrix. Each column v_i of V is an eigenvector of $A^T A$ that correspond to the eigenvalue λ_i . Note, there are n columns in V . The eigenvectors $\{v_{r+1}, \dots, v_n\}$ have 0 as the eigenvalue. That is, if $i \in \{r+1, \dots, n\}$, then $A^T A v_i = 0$.

U is also an orthogonal matrix. The first r columns of U are $u_1, \dots, u_r$, where $\sigma_i u_i = A v_i$, for $i \in \{1, \dots, r\}$. These columns are unit vectors.

$$\|u_i\|^2 = u_i^T u_i = \frac{(A v_i)^T (A v_i)}{\sigma_i \times \sigma_i} = \frac{v_i^T A^T A v_i}{\sqrt{\lambda_i} \times \sqrt{\lambda_i}} = \frac{v_i^T (\lambda_i v_i)}{\lambda_i} = v_i^T v_i = 1$$

If $i, j \in \{1, \dots, r\}$ and $i \neq j$, then u_i and u_j are orthogonal.

$$u_i^T u_j = \frac{(A v_i)^T (A v_j)}{\sigma_i \times \sigma_j} = \frac{v_i^T A^T A v_j}{\sigma_i \times \sigma_j} = \frac{v_i^T (\lambda_j v_j)}{\sigma_i \times \sigma_j} = \frac{\lambda_j (v_i^T v_j)}{\sigma_i \times \sigma_j} = \frac{\lambda_j (0)}{\sigma_i \times \sigma_j} = 0$$

The columns $u_{r+1}, \dots, u_m$ can be chosen in such a way that U is an orthogonal matrix.

Let us find AV .

$$\begin{aligned} AV &= A \begin{bmatrix} | & | & \cdots & | & | & \cdots & | \\ v_1 & v_2 & \cdots & v_r & v_{r+1} & \cdots & v_n \\ | & | & & | & | & & | \end{bmatrix} \\ &= \begin{bmatrix} | & | & \cdots & | & | & \cdots & | \\ A v_1 & A v_2 & \cdots & A v_r & A v_{r+1} & \cdots & A v_n \\ | & | & & | & | & & | \end{bmatrix} \\ &= \begin{bmatrix} | & | & \cdots & | & | & \cdots & | \\ \sigma_1 u_1 & \sigma_2 u_2 & \cdots & \sigma_r u_r & 0 & \cdots & 0 \\ | & | & & | & | & & | \end{bmatrix} \quad (1) \end{aligned}$$

The vectors $v_{r+1}, \dots, v_n$ are the eigenvectors of $A^T A$ with 0 as the eigenvalue. That is, $A^T A v_i = 0$ and $v_i \in N(A^T A)$, where $i \in \{r+1, \dots, n\}$. Recall, $N(A) = N(A^T A)$. Therefore, $v_i \in N(A)$ and $A v_i = 0$.

Let us multiply $U\Sigma$

$$\begin{aligned}
U\Sigma &= \begin{bmatrix} | & | & & | & | & & | \\ u_1 & u_2 & \cdots & u_r & u_{r+1} & \cdots & u_m \\ | & | & & | & | & & | \end{bmatrix} \times \begin{bmatrix} \sigma_1 & 0 & \cdots & 0 & 0 & \cdots & 0 \\ 0 & \sigma_2 & \cdots & 0 & 0 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \sigma_r & 0 & \cdots & 0 \\ 0 & 0 & \cdots & 0 & 0 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 0 & 0 & \cdots & 0 \end{bmatrix} \\
&= \begin{bmatrix} | & | & & | & | & & | \\ (\sigma_1)u_1 & (\sigma_2)u_2 & \cdots & (\sigma_r)u_r & (0)u_{r+1} & \cdots & (0)u_m \\ | & | & & | & | & & | \end{bmatrix} \\
&= \begin{bmatrix} | & | & & | & | & & | \\ \sigma_1 u_1 & \sigma_2 u_2 & \cdots & \sigma_r u_r & 0 & \cdots & 0 \\ | & | & & | & | & & | \end{bmatrix} \tag{2}
\end{aligned}$$

The RHS of (1) and (2) are the same. Therefore,

$$\begin{aligned}
AV &= U\Sigma \\
A &= U\Sigma V^T
\end{aligned}$$

Recall, V is orthogonal, i.e., $VV^T = I$.

All the entries in the principal diagonal of Σ , including zeros, are the **singular values** of A . The columns of U are the **left singular vectors** of A and the columns of V are the **right singular vectors** of A .

Moreover we can show that the columns of U are the eigenvectors of AA^T . The vectors $\{u_1, \dots, u_r\}$ are the eigenvectors of AA^T that correspond to $\lambda_1, \dots, \lambda_r$. The vectors $\{u_{r+1}, \dots, u_m\}$ are the eigenvectors of AA^T with 0 as the eigenvalue.

$$\begin{aligned}
AA^T &= (U\Sigma V^T)(U\Sigma V^T)^T \\
&= (U\Sigma V^T)(V\Sigma^T U^T) \\
&= U\Sigma\Sigma^T U^T
\end{aligned}$$

$\Sigma\Sigma^T$ is a $m \times m$ diagonal matrix, with the first r diagonal entries as $\lambda_1, \dots, \lambda_r$. The remaining diagonal entries are zero

$$\Sigma\Sigma^T = \begin{bmatrix} \lambda_1 & \cdots & 0 & 0 & \cdots & 0 \\ \vdots & \ddots & \vdots & \vdots & \ddots & \vdots \\ 0 & \cdots & \lambda_r & 0 & \cdots & 0 \\ 0 & \cdots & 0 & 0 & \cdots & 0 \\ \vdots & \ddots & \vdots & \vdots & \ddots & \vdots \\ 0 & \cdots & 0 & 0 & \cdots & 0 \end{bmatrix}_{m \times m}$$

4. Best Rank Approximation

Matrix $A_{m \times n}$, with $\text{Rank}(A) = r$, is decomposed as $A = U\Sigma V^T$. Here,

- U is an $m \times r$ matrix with orthonormal columns
- V is an $n \times r$ matrix with orthonormal columns
- Σ is a $r \times r$ diagonal matrix

$U\Sigma V^T$ can be written as follows:

$$\begin{bmatrix} | & | & & | \\ u_1 & u_2 & \cdots & u_r \\ | & | & & | \end{bmatrix} \times \begin{bmatrix} \sigma_1 & 0 & \cdots & 0 \\ 0 & \sigma_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \sigma_r \end{bmatrix} \times \begin{bmatrix} - & v_1^T & - \\ - & v_2^T & - \\ \vdots & \vdots & \\ - & v_r^T & - \end{bmatrix}$$

The above matrix multiplication can be expressed as the following summation,

$$A = U\Sigma V^T = \sum_{i=1}^r \sigma_i (u_i v_i^T)$$

Among all the $m \times n$ matrices with $k \leq r$ as the rank, the matrix which provides the best approximation for A is

$$\sum_{i=1}^k \sigma_i (u_i v_i^T)$$

When $k = r$, the above summation becomes equal to A .

5. Geometry Behind SVD

What happens when we multiply a vector x with a matrix A , whose rank is r ? We know that A can be decomposed as $U\Sigma V^T$.

$$Ax = U\Sigma V^T x$$

V^T is an orthogonal matrix. When we multiply $V^T x$, vector x is rotated by V^T . But, the length is preserved.

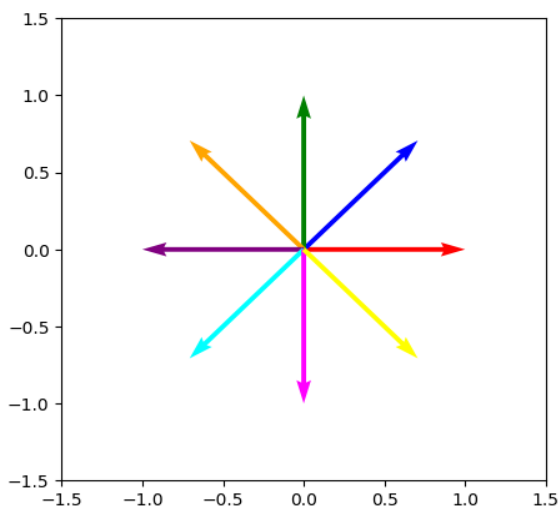
$$||V^T x|| = ||x||$$

When we multiply $\Sigma(V^T x)$, the vector $V^T x$ gets stretched or shrunk as the first r diagonal elements are greater than 0.

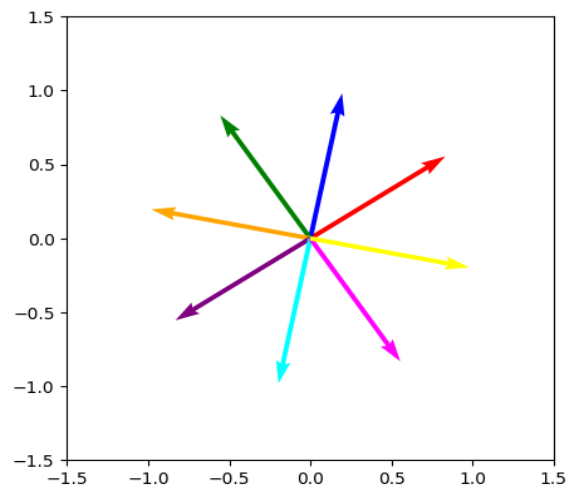
U is an orthonormal matrix. When we multiply $U(\Sigma V^T x)$, the vector $\Sigma V^T x$ is rotated. Once again, the length is preserved.

$$||U(\Sigma V^T x)|| = ||\Sigma V^T x||$$

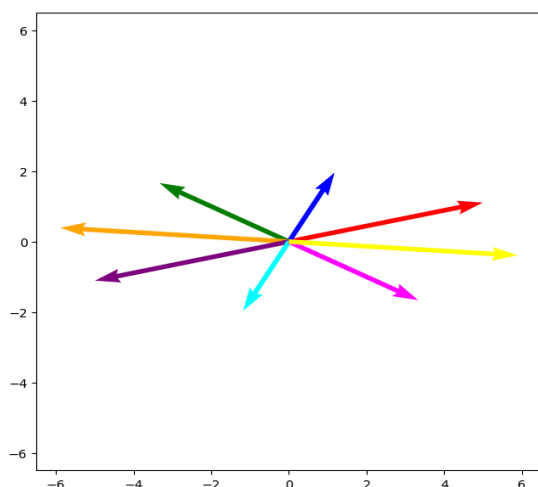
Overall, when we multiply a vector with matrix A , the vector is rotated by V^T , then stretched by Σ and is once again rotated by U .



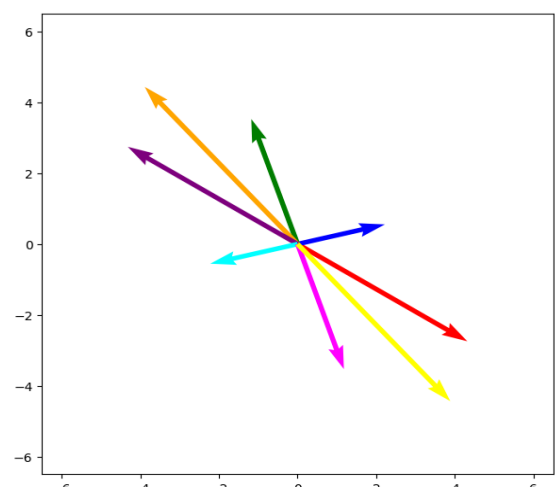
Vectors before being multiplied by V^T



Vectors after being multiplied by V^T



**Vectors after being multiplied by V^T
and then by Σ**



**Vectors after being multiplied by V^T ,
then by Σ and then by U**

Notice how the angle between the vectors is preserved after being multiplied by the orthogonal matrices V^T and U .

Just a small note: Let Q be an orthogonal matrix. There are only two possible values for $\det(Q)$, which are 1 and -1 . If $\det(Q) = 1$, then applying Q to a vector x results in a rotation. However, if $\det(Q) = -1$, then x is rotated and also reflected.

6. Definite Matrices

POSITIVE DEFINITE MATRICES

A symmetric matrix $S_{n \times n}$ is positive definite if either of the following holds true.

- All the eigenvalues of S is greater than 0
- $\forall x \in \mathbb{R}^n, x \neq 0, x^T S x > 0$

POSITIVE SEMI-DEFINITE MATRICES

A symmetric matrix $S_{n \times n}$ is positive semi-definite if either of the following holds true.

- All the eigenvalues of S is greater than or equal to 0
- $\forall x \in \mathbb{R}^n, x \neq 0, x^T S x \geq 0$

NEGATIVE DEFINITE MATRICES

A symmetric matrix $S_{n \times n}$ is negative definite if either of the following holds true.

- All the eigenvalues of S is less than 0
- $\forall x \in \mathbb{R}^n, x \neq 0, x^T S x < 0$

NEGATIVE SEMI-DEFINITE MATRICES

A symmetric matrix $S_{n \times n}$ is negative semi-definite if either of the following holds true.

- All the eigenvalues of S is less than or equal to 0
- $\forall x \in \mathbb{R}^n, x \neq 0, x^T S x \leq 0$

INDEFINITE MATRICES

A symmetric matrix is **indefinite** if it is neither positive definite nor positive semi-definite nor negative definite nor negative semi-definite.

7. Definite Functions

Consider $f : \mathbb{R}^d \rightarrow \mathbb{R}, d \geq 1$.

POSITIVE DEFINITE FUNCTIONS

→ f is positive definite if

- $f(0) = 0$
- $f(x) > 0$ if $x \in \mathbb{R}^d$ and $x \neq 0$

→ 0 is the only global minimum of f

POSITIVE SEMI-DEFINITE FUNCTIONS

→ f is positive semi-definite if

- $f(x) \geq 0, \forall x \in \mathbb{R}^d$

→ 0 is a global minimum of f , but not necessarily unique

NEGATIVE DEFINITE FUNCTIONS

→ f is negative definite if

- $f(0) = 0$
- $f(x) < 0$ if $x \in \mathbb{R}^d$ and $x \neq 0$

→ 0 is the only global maximum of f

NEGATIVE SEMI-DEFINITE FUNCTIONS

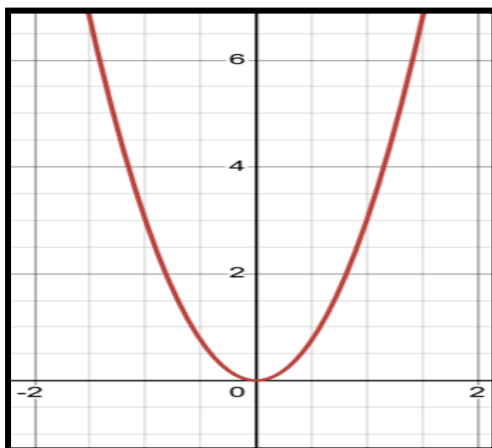
→ f is negative semi-definite if

- $f(x) \leq 0, \forall x \in \mathbb{R}^d$

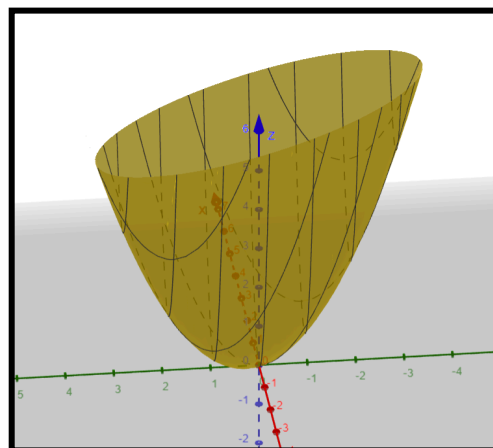
→ 0 is a global maximum of f , but not necessarily unique

EXAMPLES OF POSITIVE DEFINITE FUNCTIONS

- $f(x) = kx^2$, where $k > 0$
- $f(x, y) = ax^2 + 2bxy + cy^2$, if $a > 0, c > 0$ and $ac - b^2 > 0$. We shall prove this later.



$$f(x) = 3x^2$$



$$f(x) = 2x^2 + y^2 + 2xy$$

For $f(x, y) = ax^2 + 2bxy + cy^2$, we have to show that f is positive definite if

$$a > 0, c > 0 \text{ and } ac - b^2 > 0.$$

First, $f(0, 0) = 0$. $f(1, 0)$ and $f(0, 1)$ should be greater than 0

$$f(1, 0) > 0$$

$$a > 0$$

$$f(0, 1) > 0$$

$$c > 0$$

If $f(x, y) = ax^2 + 2bxy + cy^2$ has to be greater than zero when $(x, y) \neq (0, 0)$, then $a > 0$ and $c > 0$.

Let us now focus on b .

$$\begin{aligned} f(x, y) &= ax^2 + 2bxy + cy^2 \\ &= a \left[x^2 + \frac{2b}{a}xy \right] + cy^2 \\ &= a \left[x^2 + 2x \left(\frac{by}{a} \right) + \left(\frac{by}{a} \right)^2 - \left(\frac{by}{a} \right)^2 \right] + cy^2 \\ &= a \left[x^2 + 2x \left(\frac{by}{a} \right) + \left(\frac{by}{a} \right)^2 \right] - \frac{b^2y^2}{a} + cy^2 \\ &= a \left[x + \frac{by}{a} \right]^2 + y^2 \left[c - \frac{b^2}{a} \right] \end{aligned}$$

$f(x, y)$ is a sum of two squared terms. One term is $u = \left[x + \frac{by}{a} \right]^2$, the other term is $v = y^2$.

$f(x, y)$ will be greater than 0 whenever $(x, y) \neq (0, 0)$ if the coefficients of u and v are greater than 0.

- $a > 0$
- $c - \frac{b^2}{a} > 0$

$$\begin{aligned} c - \frac{b^2}{a} &> 0 \\ c &> \frac{b^2}{a} \\ ac - b^2 &> 0 \end{aligned}$$

RELATIONSHIP BETWEEN DEFINITE FUNCTIONS AND DEFINITE MATRICES

The function $f(x, y) = ax^2 + 2bxy + cy^2$ can be written as

$$f(x, y) = ax^2 + 2bxy + cy^2 = \begin{bmatrix} x & y \end{bmatrix} \begin{bmatrix} a & b \\ b & c \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$$

Matrix $S = \begin{bmatrix} a & b \\ b & c \end{bmatrix}$ is a symmetric matrix. The function $f(x, y) = ax^2 + 2bxy + cy^2$ is :

- positive definite if S is positive definite
- positive semi-definite if S is positive semi-definite
- negative definite if S is negative definite
- negative semi-definite if S is negative semi-definite

This is true for any function $f : \mathbb{R}^d \rightarrow \mathbb{R}$, that can be expressed as

$$f(x) = x^T S x$$

where S is a symmetric matrix.
